12.12 When to Stop Looking

WORKSHEET

The question

STEP I 2001, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q14

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself In a random order, what is the probability that the overall best candidate is among the first n interviewed?

STAGE 2 Best or second best

Ask yourself The best of the first n is the second best overall exactly when the second best is among the first n but the best is not. What is that probability?

STAGE 3 The optimal strategy

Ask yourself Strategy: reject the first r candidates, then accept the first candidate better than all of them. If the best candidate is in position $k > r$, when does the strategy pick them?

Ask yourself For large N, write $x = \frac{r}{N}$. The sum is approximately $\ln \frac{N}{r}$. What value of x maximises $x \ln \frac{1}{x}$?

Try this next

1. **Ice-creams.** Try STEP II 2006, Question 13 (step.maths.org/questions/06-s2-q13): a small version of the secretary problem with a simple rule. Reject the first, then take the first that beats it.

2. **Small N.** For $N = 4$, find the best value of r and the probability of success. Check by listing.

3. **Best or second best.** Using part (ii), argue that "the best of the first n is one of the top two" is much more likely than "the best of the first n is the best". What does this suggest about strategies for getting one of the top two?

4. **An exact maximum.** For $N = 10$, check from the figure that $r = 3$ is optimal. For $N = 100$, show that $r = 37$ is optimal (to four decimal places the success probabilities for $r = 36, 37, 38$ are 0.3710, 0.3710, 0.3708).

Further reading

► **Thomas Ferguson, "Who solved the secretary problem?"**, *Statistical Science* 4 (1989).

History and variations; free online.

- **Brian Christian & Tom Griffiths, *Algorithms to Live By*** (William Collins). Optimal stopping and other algorithms applied to everyday decisions.
- **Martin Gardner, *New Mathematical Diversions*** (Mathematical Association of America). The game of googol.

The full guided solution, and what the question leads to at university, is in Section 12.12:
https://beyondstep.keithmcnulty.org/ch12/secretary.html